TECHNICAL REPORT

Customer Churn Prediction & Revenue Protection Strategy

1. Introduction

1.1 Business Context

Customer churn is a major revenue risk in subscription-based businesses. Since acquiring new customers is significantly more expensive than retaining existing ones, early churn prediction is critical for financial stability.

This project analyzes telecom data (500 customers) to identify churn drivers, build predictive models, estimate revenue at risk, and recommend targeted retention strategies.

The overall churn rate observed in the dataset is **10.60%**, indicating a measurable but manageable business risk

1.2 Business Objectives

- To reduce churn rate using predictive analytics
- To proactively identify high-risk customers
- To improve customer lifetime value (CLV)
- To protect recurring annual revenue

1.3 Key Performance Indicators (KPIs)

➢ **Overall Churn Rate (10.60%)**
Baseline business risk derived from EDA.

➢ **Model Recall (Churn Class = 0.69 – 0.88)**

- Logistic Regression Recall: **0.88**
- Random Forest Recall: **0.69**
  Ensures churners are correctly identified before leaving.

➢ **ROC-AUC Score (0.9926)**
Indicates excellent model discrimination capability.
Cross-validation ROC-AUC: **0.9787**, confirming model stability.

- **Revenue at Risk (₹10,212 Estimated)**
  Calculated for customers with churn probability > 0.6, using:

  $$\text{MonthlyCharges} \times 12.$$

- **Precision for High-Risk Segment (Up to 1.00)**
  Random Forest achieved perfect precision for churn class in test data, ensuring retention campaigns are highly targeted and cost-efficient.

2. Data Description

2.1 Dataset Overview

Dataset: customer_churn.csv
Total Records: 500 customers
Total Features: 9 original features + 1 engineered feature

Target Variable: Churn (0 = Retained, 1 = Churned)

Data Quality Summary:

- No missing values detected
- No duplicate records
- Final dataset shape after cleaning: **(500, 10)**

Feature Summary

Feature	Description
CustomerID	Unique customer identifier
Tenure	Number of months customer stayed
MonthlyCharges	Monthly subscription cost
TotalCharges	Total lifetime revenue from customer
Contract	Contract type (Month-to-month, One year, Two year)
PaymentMethod	Mode of payment
PaperlessBilling	Billing preference (Yes/No)

Feature	Description
SeniorCitizen	Age category indicator (0/1)
Churn	Target variable
AvgMonthlySpend	Engineered feature (TotalCharges / Tenure)

2.2 Data Types

- **Numerical Features:**
  Tenure, MonthlyCharges, TotalCharges, AvgMonthlySpend
- **Categorical Features:**
  Contract, PaymentMethod, PaperlessBilling
- **Binary Features:**
  SeniorCitizen, Churn

Observations

- Strong statistical relationship found between Tenure and Churn (p-value ≈ 2.43e-34).
- Significant association between Contract Type and Churn (Chi-square p-value ≈ 9.59e-07).
- Dataset is clean and analysis-ready without major preprocessing constraints.

This dataset is suitable for predictive modeling and business-driven churn risk analysis.

3. Data Cleaning Process

3.1 Missing Values Handling

- Converted TotalCharges to numeric format for consistency.
- Verified dataset completeness — no missing values were found (0 nulls across all columns).

Rationale: Ensuring correct data types is critical for statistical testing and machine learning compatibility.

3.2 Duplicate Removal

- Checked and removed duplicate records.
- Final dataset size remained **500 rows**, confirming no duplicate customer entries were present.

3.3 Feature Engineering

Created new feature:

AvgMonthlySpend = TotalCharges / (Tenure + 1)

Purpose:

- Capture spending intensity relative to customer lifespan
- Reduce bias from high TotalCharges in long-tenure customers

3.4 Data Validation

- Confirmed dataset meets minimum size requirement (500 rows)
- Verified 0 missing values post-cleaning
- Final dataset shape: **(500, 10)**

4. Exploratory Data Analysis (EDA)

4.1 Churn Distribution

Observed churn rate: 10.60%

Insight:
The dataset is moderately imbalanced (approx. 89% retained vs 11% churned) but still suitable for modeling without resampling techniques.

4.2 Tenure vs Churn

Finding: T-Test p-value = **2.43e-34** (highly significant)
Churned customers have significantly lower tenure compared to retained customers.

Business Meaning:
Customer attrition is concentrated in the early lifecycle stage. Onboarding and early engagement strategies are critical.

4.3 Monthly Charges vs Churn

Finding: MonthlyCharges shows a positive correlation with churn (confirmed in heatmap).

Business Meaning:
Customers with higher monthly bills are more price-sensitive and more likely to churn. Pricing optimization or bundled offers may reduce risk.

4.4 Contract Type vs Churn

Finding: Chi-Square p-value = **9.58e-07** (statistically significant)
Month-to-month contracts exhibit the highest churn rate.

Business Meaning:
Lack of contractual commitment increases volatility. Long-term contract conversion strategies can reduce churn.

4.5 Correlation Analysis

Key Correlations Identified:

- Tenure → Negative correlation with churn
- MonthlyCharges → Moderate positive correlation with churn

Interpretation:
Customer loyalty (tenure) reduces churn probability, while higher recurring costs increase churn likelihood.

5. Statistical Testing

5.1 T-Test (Tenure Difference)

Objective: Test whether average tenure differs between churned and retained customers.

Result: p-value = 2.43e-34 ($\ll 0.05$)

Conclusion:
There is a highly statistically significant difference in tenure between churned and retained customers.
Customers with lower tenure are significantly more likely to churn.

5.2 Chi-Square Test (Contract vs Churn)

Objective: Test whether contract type and churn are dependent.

Result: p-value = 9.59e-07 (<< 0.05)

Conclusion:
Contract type has a strong statistically significant association with churn. Month-to-month contracts contribute disproportionately to churn risk.

6. Model Development

6.1 Preprocessing

- Label encoding applied to categorical features
- Standard scaling applied to all numerical features
- Train-test split performed (80/20)
- random_state = 42 used for reproducibility

Dataset distribution:

- Total records: 500
- Test set size: 100
- Churn rate: **10.6%** (imbalanced dataset)

6.2 Models Implemented

1. Logistic Regression

- Used as baseline interpretable model
- Suitable for churn probability estimation
- Test Accuracy: 94%
- Recall (Churn class): 88%
- Strong performance for minority class detection

2. Random Forest (Tuned Model)

- Ensemble-based model capturing nonlinear patterns
- Hyperparameter tuning via GridSearchCV
- Best Parameters:

- n_estimators = 100
- max_depth = 5

- Test Accuracy: **95%**
- Recall (Churn class): **69%**
- ROC-AUC: **0.9926**

6.3 Cross-Validation

- 5-fold cross-validation used
- Evaluation metric: ROC-AUC
- Mean ROC-AUC: **0.9787**

Purpose:

- Validate model stability
- Ensure strong generalization
- Reduce overfitting risk

6.4 Hyperparameter Tuning

GridSearchCV optimized:

- Number of estimators
- Maximum tree depth

Best model selected based on highest ROC-AUC score.

7. Model Evaluation

7.1 Classification Metrics

Model performance was evaluated using:

- Precision
- Recall
- F1-score
- ROC-AUC

Best Model: Random Forest (Tuned)

Key Results (Test Set):

- Accuracy: **95%**
- Recall (Churn class): **69%**
- Precision (Churn class): **100%**
- ROC-AUC: **0.99**

Interpretation:
The model identifies churners with high precision (no false positives), but recall indicates some churners are still missed. Since churn prevention prioritizes minimizing false negatives, recall can be further optimized if needed.

7.2 ROC Curve Analysis

- ROC-AUC Score: **0.9925**
- Cross-Validation ROC-AUC: **0.9787**

Interpretation:
The ROC curve demonstrates excellent separation capability. The model strongly distinguishes between churned and retained customers with very high discriminative power.

7.3 Feature Importance

Top predictors identified:

- Tenure
- Contract Type
- Monthly Charges
- AvgMonthlySpend

Interpretation:
Customer behavior (tenure) and pricing structure (charges and contract type) are the strongest drivers of churn. Short-tenure, high-charge, month-to-month customers are at highest risk.

8. Revenue Impact Analysis

8.1 Revenue Assumption

Annual Revenue per Customer = MonthlyCharges × 12

Customers with churn probability > 0.60 classified as high-risk segment

8.2 Revenue at Risk

Model (ROC-AUC: 0.99) identifies high-risk customers representing ₹10,212 annual revenue at risk.

Business Insight:
Targeting customers with churn probability > 0.60 can directly protect this revenue and generate measurable retention impact.

9. Business Translation

9.1 Key Drivers Identified (Statistically Validated)

- Short tenure (T-test p-value = 2.43e-34 → highly significant)
- Month-to-month contracts
- High monthly charges
- Payment method impact (notably electronic payments)

9.2 Recommended Actions

1. Offer discounts to convert month-to-month customers to annual contracts
2. Launch onboarding engagement for customers with tenure < 12 months
3. Introduce pricing bundles for high monthly charge segments
4. Deploy churn scoring model into CRM for proactive retention

9.3 Implementation Roadmap

Phase	Action	Timeline
Phase 1	Deploy trained model	Day 1–2

Phase	Action	Timeline
Phase 2	Identify high-risk customers (>0.6 probability)	Day 3
Phase 3	Launch targeted retention campaign	Day 4–5
Phase 4	Track churn rate & revenue impact	Day 6–7

10. Limitations

- Dataset size limited to 500 records
- No detailed behavioral or usage data
- Simplified revenue estimation
- No time-based churn tracking

11. Future Improvements

1. Integrate usage and engagement metrics
2. Apply advanced models (e.g., gradient boosting)
3. Add model explainability (feature-level interpretation)
4. Use time-based validation strategy
5. Implement uplift modeling for campaign targeting
6. Automate pipeline deployment

12. Conclusion

This project delivers a validated end-to-end churn prediction system with strong predictive performance (ROC-AUC 0.99) and quantified revenue exposure. The model enables proactive retention, data-driven targeting, and measurable business impact.